

RAMAN ARUN RANDIVE

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Summary

Computer Engineering student specializing in **Production AI Systems** and **Cloud Infrastructure**. Expert in architecting **RAG pipelines**, semantic search, and scalable backends using Python, LangChain, and PyTorch. Managed technical delivery for 5+ initiatives and optimized deep learning models for real-world deployment.

Technical Skills

AI & Machine Learning: Deep Learning, NLP, RAG, Prompt Engineering, Vector Search, Semantic Search, LangChain, HuggingFace, PyTorch

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Linux, Git, REST APIs, WebRTC, Prisma, Supabase

Frameworks & Databases: FastAPI, Node.js, Express, React.js, Next.js 15, PostgreSQL, MongoDB, Pinecone, Neo4j

Languages: Python, TypeScript, JavaScript, C++, SQL, Cypher

Experience

CESA ACM Student Chapter, PCCOE

Pune, India

Technical Head

Aug 2025 – Present

- Spearheaded technical architecture for **5+ AI/ML initiatives**, increasing deployment efficiency by **30%** and reducing server downtime by **15%**.
- Managed and mentored **50+ students** on System Design and DSA, resulting in a **20% increase** in national-level contest qualifications.
- Directed development of internal tools using React and FastAPI to digitize faculty Appraisal System for **40+ active Faculties**.

Technical Projects

Enterprise RAG Pipeline & Chatbot | *Next.js 15, FastAPI, Pinecone, Neo4j* | [Live Demo](#)

Oct 2025

- Engineered a high-performance RAG chatbot that **reduced retrieval latency by 40%** and improved query accuracy by **25%**.
- Developed secure thread-based interactions using Supabase Auth and JWT, enabling concurrent processing of **multiple large-scale datasets**.
- Optimized context-aware responses by integrating **Named Entity Recognition (NER)** for Chat system, resulting in a **30% improvement** in precision.

RL-Based Network Intrusion System | *Reinforcement Learning, Distributed Systems* | [Live Demo](#)

Sept 2025

- Architected an automated environment for SRE incident diagnosis, simulating BGP/DNS faults to improve AI agent decision-making by **15%**.
- Built a scalable reward engine to benchmark agents across **3 distinct task domains**, reducing evaluation time by **20%**.
- Provisioned **5+ API endpoints** to facilitate real-time agent-environment interaction within a distributed network infrastructure.

Research & Publications

Spatiotemporal CNNs for Video Deepfake Detection(Under Review)

Aug 2025

- Developed a 3D ResNet-18 model for feature extraction on FaceForensics++, achieving a state-of-the-art **AUC of 0.9344**.
- Published a survey paper on **GATv2 Graph Neural Networks** and secured an Indian Copyright for geospatial predictive routing.

Education

Pimpri Chinchwad College of Engineering (PCCOE)

Pune, India

B.Tech in Computer Engineering (CGPA: 8.35 / 10.0)

Aug 2023 – Jun 2027

- **Multidisciplinary Minor:** Generative AI

Certifications & Achievements

- **Harvard CS50x & CS50 AI:** Certified in AI foundations and Neural Networks.
- **Finalist in 3+ National Hackathons :** Cavista 2026, VIT Pune 2025, TE-Connectivity AI CUP.
- **Competitive Programming:** Solved **400+ problems** across Codeforces, LeetCode, and CodeChef.
- **Explainable AI (XAI):** Completed specialized training at the ACM India Winter School on model interpretability.